Features of the circle

1

a Radius **b** Diameter **c** Circumference **d** Centre

2

a 7 cm **b** 5 cm

3 $2x$

4 40 cm

5

a

Diameter (cm)	Radius (cm)
22	11
34	17
38	19
42	21
46	23

b

Diameter (cm)	Radius (cm)
22	11
26	13
28	14
36	18
42	21

Circumference

6

a False **b** True **c** True **d** False

e False **f** False

7

a

i $13\pi\text{ cm}$ **ii** 40.84 cm

b

i $36\pi\text{ cm}$ **ii** 113.10 cm

c

i $7.7\pi\text{ cm}$ **ii** 24.19 cm

d

i $8.2\pi\text{ cm}$ **ii** 25.76 cm

8

a $59\pi\text{ cm}$ **b** $44\pi\text{ cm}$

9

a 119.38 cm **b** 141.37 cm **c** 169.65 cm **d** 11.31 cm

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10

a 10π cm

b 10π cm

c They got the same result because the diameter is always equal to twice the radius, so πd and $2\pi r$ are always equal.

11

a $\frac{26}{\pi}$ cm

b 8.28 cm

12

a 6 cm

b $\frac{6}{\pi}$ cm

c 16 cm

d $\frac{13}{\pi}$ cm

13

a 2.23 cm

b 2.86 cm

c 2.37 cm

d 15.60 cm

14

a $\frac{36}{\pi}$ cm

b 11.46 cm

15

a 18 cm

b $\frac{18}{\pi}$ cm

c 17 cm

d $\frac{14}{\pi}$ cm

16

a 11.78 cm

b 14.01 cm

c 22.28 cm

d 7.80 cm

17

Radius (cm)	Diameter (cm)	Circumference (cm)
2	4	4π
8	16	16π
41	82	82π
$\frac{170}{\pi}$	$\frac{340}{\pi}$	340

18

a 13.37 m

b 6.68 m

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Applications

19 100.5 cm

20 106.8 cm

21 25.1 m

22 94.2 mm

23 311.0 cm

24 22.1 laps

25 39 788 revolutions

26 345.6 m

27 163.4 m

28 8

29 879.6 m

30 231.0 km

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